

The Referee Is the Product:

Un-gameable Verification for LLM-Generated GPU Kernels, and What It Falsified About Its Builders’ Beliefs

Rohit Mahendra Tigon

YMRohit artifacts: huggingface.co/YMRohit/ouroboros-kernelsmith-qwen3.6-27b

June 2026 (preprint; single-seed ablations labeled as such)

Abstract

Large-language-model kernel generation has a credibility problem: the most publicized result in the area was retracted within days when its generated kernels were found to be exploiting the evaluation harness rather than the hardware. We argue the root cause is architectural — the reward channel is treated as a metric when it must be engineered as an *adversarial system under test* — and we present OUROBOROS, a verifier-grounded self-distillation loop whose central artifact is an immutable referee with *negative controls for the reward function itself*: implemented exploits (benchmark-shape special-casing, pointer-keyed output memoization, in-place input mutation) ship in the test suite and must be REJECTED for the system to be considered sound. Trained only by this referee, a 27B-parameter open model (Qwen3.6-27B, LoRA) wrote 76 Triton fusion kernels that *all* beat `torch.compile` max-autotune reproducibly: 69/69 pass a $5\times$ fresh-run stability gate (worst kernel $1.113\times\pm0.015$; best $2.038\times\pm0.031$), and every kernel holds a geometric-mean win across a $(M, N)\times\text{dtype}$ shape grid of 816 cells with per-cell recompiled baselines and cache-cold verification — with the 10.9% of cells we lose reported individually. Ablations show that on familiar operators a frozen model plus verified best-of- N search captures 97% of the full loop’s value and the policy-gradient term earns nothing, while on never-seen operators learning is visibly load-bearing (validity climbing from $\sim 50\%$ to 68% within one run as the model distills its own verified solutions). Finally, we document nine instances in which the verifier falsified a confident belief held by the system’s own builders — including our published diagnosis of the system’s one weak regime, which the model repaired with a simpler schedule family than the one we predicted, flipping the worst published loss ($0.69\times$) into a $1.222\times$ win. We conclude that in domains with cheap, honest measurement, the scarce engineering asset is not the model but the referee, and we release the harness, all 76 kernels, the corpus, adapters, raw per-sample measurements, and a falsification ledger.

1 Introduction

In February 2025, a well-funded research lab announced an “AI CUDA Engineer” with claimed speedups of up to $150\times$; within roughly 48 hours, public reviewers found the generated kernels were reusing memory across evaluation runs — gaming the benchmark harness, not accelerating the hardware — and the headline numbers collapsed [1]. The episode is usually told as a cautionary tale about hype. We read it differently: it was a *systems failure with a systems fix*. The reward channel of an RL-for-code pipeline is an attack surface, and like any attack surface it can be hardened, regression-tested, and shipped with its known exploits as test cases.

This paper presents OUROBOROS, a small worked system built around that principle, and reports what happened when we ran it for one sustained campaign. Concretely:

1. **The harness is the product (§3).** An immutable referee compiles each candidate Triton kernel, checks elementwise correctness against a PyTorch reference across *adversarial* shapes, dtypes, and magnitudes, and measures wall-clock with CUDA events under a measurement-hygiene stack (§3.2). Crucially, the referee ships with *negative controls for its own reward channel*: three implemented cheat kernels — each a working exploit against a naive version of the harness — that must be REJECTED for the self-test to pass. The full self-test is 36 cases (17 gold kernels must pass, 19 negative controls must fail) and is ALL GREEN on both an RTX 4090 and an H200.

2. **The loop** (§4): supervised bootstrap to a measured validity gate, then verifier-rewarded search with self-distillation on the model’s own verified winners (reward = measured speedup; Eq. 2), with dedup by canonicalized AST hash and winner’s-curse-corrected exit validation.
3. **Results** (§5): 76 model-era kernels; 69/69 reproducible discoveries against `torch.compile` max-autotune (the incumbent autotuner, i.e., the strongest honest compiler baseline); geomean wins across the entire shape grid with losses (89/816 cells) reported per-cell and concentrated in one explainable regime; parity-or-better against expert hand-written Triton under *two* comparison conditions including the library-as-shipped one; and a correctness-gated end-to-end transformer-block composition with its Amdahl ceiling stated.
4. **Ablations and the locus of learning** (§6): on familiar operators, frozen best-of- N against the referee captures 97% of the full loop’s geomean and removing the policy-gradient term does not hurt (single-seed; labeled). On foreign operators, learning is load-bearing and visible within a single run (Fig. 3).
5. **The falsification ledger** (§7): nine concrete instances where the referee overruled a confident human belief — including the authors’ own published diagnosis, hand-written “textbook” kernel, ablation prediction, and one of our own earlier claims, which we retract here explicitly.

We do *not* claim novel algorithm discovery (§8); the wins are reproducible *scheduling* wins of ~ 10 –100% on bandwidth-bound fusion operators, which is precisely the long tail practitioners hand-write today. The claim we defend is methodological: **with an un-gameable referee, a commodity open model out-searches the incumbent autotuner, repairs its own weaknesses, and corrects its builders — and every one of those statements is regenerable from a machine-emitted artifact in our repository.** A number that cannot be regenerated by a command does not appear in this paper; every statistic below is compiled from the harness’s JSON outputs by the same script that performs cross-report consistency checks.

2 Related Work

LLMs writing GPU kernels. KernelBench [2] established the evaluation regime; entrants include Kevin-32B (multi-turn RL) [3], AutoTriton (RL for Triton) [4], CUDA-L1 (contrastive RL) [5], and the AI CUDA Engineer [1]. Our work differs in locus: we contribute the *verification system* and treat the model as a replaceable commodity — our pipeline ran unchanged across a 2B and a 27B model. We also intentionally evaluate against the compiler’s own autotuner rather than eager PyTorch, since eager baselines flatter fusion work.

Machine-discovered algorithms and superoptimization. AlphaTensor [6] and AlphaDev [7] demonstrated the prize; classical superoptimization (e.g., STOKE [8]) established search-with-verification. OUROBOROS is the commodity-model, cheap-oracle version of this recipe, with the oracle-hardening itself as the contribution.

Reward hacking and verifiable rewards. RL from verifiable rewards is now standard in frontier training [9, 10]; reward gaming is its known failure mode. Our proposal — ship the reward function with implemented exploits as regression tests (“negative controls for rewards”) — is, to our knowledge, not systematically practiced anywhere in the RL-for-code literature, despite being cheap and decisive.

Compilers. `torch.compile`/Inductor with `max-autotune` [11] is our baseline *and* the incumbent we measure against; Triton [12] is the target language; Liger-Kernel [13] provides the expert hand-written comparison set.

3 The Referee

A candidate kernel is a Python source string defining `run(*inputs)` and its `@triton.jit` kernel(s). The referee (`harness.py`, ~ 370 lines, subprocess-isolated with hard timeouts) is the only component that ever executes one, and returns a verdict that neither the proposer nor any trainer can influence:

storage dtype	rtol	atol
fp32	2×10^{-4}	2×10^{-5}
fp16	2×10^{-2}	2×10^{-3}
bf16	3×10^{-2}	8×10^{-3}

Table 1: Derived tolerance envelopes (internals accumulate in fp32). Per-operator overrides exist only where derived and documented (scan; entropy).

$$\text{verdict}(k) = \left(\text{CORRECT}(k) \in \{0, 1\}, \quad s_b(k) = \frac{\text{median}_{i \leq N} t_b^{(i)}}{\text{median}_{i \leq N} t_k^{(i)}} \right) \quad (1)$$

where b ranges over baselines (eager; `torch.compile` default; `torch.compile` max-autotune) and timings are CUDA-event medians ($N=100$ in-search, $N=50$ on grids).

3.1 Adversarial correctness

`CORRECT(k)` requires `allclose` against the PyTorch reference on *every* case in a sweep designed to make plausible-but-wrong kernels fail deterministically: (i) fixed stress cases at large magnitude ($\times 64$) in fp16/bf16 with non-power-of-two row lengths (e.g. 37×4097) — these kill the classic traps (softmax without max-subtraction, RMSNorm without `rsqrt`, dropped residuals); (ii) **the benchmark inputs themselves** (§3.3); and (iii) k randomized shape/dtype/magnitude draws. Tolerances are *derived*, not tuned: for storage dtype with unit roundoff ε , reductions over D elements accumulate $O(\sqrt{D}\varepsilon)$ relative error in fp32, giving the per-dtype envelopes (rtol, atol) of Table 1. Where the global envelope is the wrong yardstick the override is derived at the spec and documented: for a length- N prefix scan the error tracks the *running-path magnitude* $\sigma\sqrt{j}$, giving $\text{atol} \approx c\varepsilon_{\text{fp32}}\sigma_{\max}\sqrt{N_{\max}}$; for entropy-from-logits the *reference itself* carries the cancellation $H = \text{lse}(x) - \sum_i x_i p_i$, bounding attainable agreement for *any* correct kernel. One candidate operator (`l1norm`) was *excluded* on these grounds: its outputs sit below the fp16 atol, so an all-zeros kernel would pass — an unverifiable spec is not allowed to exist in the suite.

3.2 Measurement hygiene

Every guard below exists because its absence produced (or would produce) a specific, recorded false result: **(a)** CUDA events with device synchronization — wall-clock around an async launch is meaningless. **(b)** Warmup of *both* paths (25 iters) — otherwise one times JIT/Inductor compilation. **(c)** Median-of- N — boost clocks drift. **(d)** Clone/setup *outside* the timed window — an early version cloned inputs inside it, adding a $\sim 134\text{MB}$ memcpy to every measurement and pulling all ratios toward 1.0 (manufacturing “ties” with the compiler). **(e)** Winner’s-curse correction: the search archives the *max* over noisy measurements, a biased estimator; published numbers come from a fresh exit validation, and headline claims from the $5 \times$ stability gate of §5. **(f)** An opt-in *rotating-buffer* mode cycles 4 input clone-sets inside the timed loop (for candidate and baselines alike) so nothing stays L2-resident — the cache-cold cross-check.

3.3 Negative controls for the reward channel

The distinctive design element. We wrote three working exploits against the naive harness and committed them as permanent test cases:

1. **Bench-shape special-casing.** The timing inputs are fixed and public; a kernel that returns garbage instantly at exactly that shape would have passed the v1 correctness sweep (whose random generator could not produce the bench shape) and posted an arbitrarily large “speedup.” *Countermeasure:* the bench inputs are part of the correctness sweep, at every grid cell.
2. **Pointer-keyed memoization.** A `run()` that caches its output by input `data_ptr` does no GPU work after warmup. *Countermeasure:* before every timed iteration (outside the event window) one input

element is overwritten with a *large in-distribution* value, and the final timed output is `allclose`-verified against the reference on the live input state. The first version of this poke — copying a neighboring element — was itself defeated by the control (the perturbation moved softmax outputs by $\sim 10^{-4}$, inside fp16 tolerance); the control caught the guard’s weakness before any model could.

3. **In-place input mutation.** Correct values written into the input tensor make the timed loop’s identical-arguments assumption unsound. *Countermeasure:* the correctness phase compares inputs before/after and rejects mutators outright.

All three must be REJECTED — each by its intended guard, with its specific error — for the self-test to pass. They are, on both GPUs tested (36/36 on NVIDIA GeForce RTX 4090; 36/36 on NVIDIA H200). We commend this practice — *unit tests for reward functions* — to anyone training against measured rewards.

4 The Loop

4.1 Operator suite

101 operator specifications: 16 explicit ops (norms, softmax family, GLU family, RoPE variants, int8 dequant, fused cross-entropy), a generative fusion grammar $[\pm \text{residual}] \rightarrow \{\text{rms, layer}\} \text{norm} \rightarrow \text{epilogue}$ over 19 epilogues (76 chain ops, each verified with two gold template variants and cross-epilogue negative controls), and regime/algorithm-class probes (§5.5). An operator enters the loop only after its gold seed passes the referee *and* its negative control is rejected.

4.2 Phase 1: supervised bootstrap to a measured gate

A cold model emits 0% valid Triton. We SFT a LoRA on a *structurally diverse*, harness-filtered corpus (teacher-written implementations $\times$ launch-knob variants; only verified survivors enter), and gate on measured validity: sampled at the RL temperatures, per-op valid-rate $\geq 80\%$ with > 1 distinct structure (the recorded run reached 100%). Knob-twiddled copies of a few seeds teach “memorize N programs”; structural diversity is what teaches “write Triton.”

4.3 Phase 2: verifier-rewarded search with self-distillation

Each round, for operator c : sample a group $\{k_i\}_{i=1}^G$ from $\pi_\theta(\cdot | c, \text{feedback})$, evaluate each through the referee (deduplicated by canonicalized-AST hash so a cosmetic rewrite is never re-measured), and assign

$$R(k) = \begin{cases} 1 + s_{\text{compile}}(k) & \text{CORRECT}(k) \\ 0 & \text{incorrect} \\ -0.2 & \text{runtime fail / timeout / crash} \\ -0.5 & \text{compile fail} \end{cases} \quad (2)$$

(correctness first, then speed — one scalar). The update combines a group-relative policy gradient [9] with self-distillation on the round’s fastest verified kernel k^* :

$$\mathcal{L} = -\frac{1}{G} \sum_{i=1}^G A_i \log \pi_\theta(k_i | c) - \log \pi_\theta(k^* | c) + \beta |\log \pi_\theta(k_i | c) - \log \pi_{\text{ref}}(k_i | c)|, \quad A_i = \frac{R_i - \bar{R}}{\sigma_R} \quad (3)$$

where the third term is a *drift penalty* to a frozen reference — honestly labeled: it is an absolute sequence-log-probability difference, a crude one-sample KL proxy, and the production runs disable it ($\beta=0$) without observed collapse (§6 suggests why: self-distillation, not the gradient term, carries the learning). Prompts carry the referee’s structured error feedback from the previous round (the fix-then-retry reflex). The archive keeps one champion per operator; exit validation re-measures every champion fresh with the `torch.compile` max-autotune baseline.

5 Results

All numbers in this section are emitted by the referee and compiled into this PDF by `make_numbers.py`, which also enforces cross-report consistency (12/12 checks at build time). Hardware: NVIDIA H200; torch 2.10; Triton 3.6; model Qwen3.6-27B (LoRA rank 128) unless stated.

5.1 The product: 76 kernels, all reproducible discoveries

Stability gate. Every archived kernel re-benchmarked $5\times$ fresh (full referee, anti-gaming guards live, `torch.compile` max-autotune recompiled each run). Discovery bar: $\bar{s} - \hat{\sigma} > 1$ where $\hat{\sigma}$ is the population spread of the five runs. Result: **69/69 discoveries; zero failures; zero marginals** (Fig. 2). Mean of per-op means $1.299\times$; maximum spread ± 0.052 ; weakest `add_layernorm_gelu_erf` at $1.113 \times \pm 0.015$; strongest `cross_entropy` at $2.038 \times \pm 0.031$ (model-authored).

Shape generality. Every kernel was re-benchmarked across a $(M, N) \times \{\text{fp16, bf16}\}$ grid (matrix family: $1024 \times 4096 \dots 256 \times 16384$; rope family over (M, D)), with `torch.compile` max-autotune *recompiled per cell* and the anti-gaming guards active in every cell — 816 cells total. **All 69 kernels hold geomean > 1** (overall $1.494\times$ on the 32 v1-era kernels, $1.480\times$ on the 37 discovery-run kernels), all survive the cache-cold rotated check (69/69), and 89 cells (10.9%) are *losses, reported individually* (Fig. 1). The losses have structure: 76 of 89 concentrate at 16384×2048 — many short rows — across two independently trained kernel generations. §5.5 returns to this regime.

5.2 Against expert hand-written Triton: two conditions

Comparisons against Liger-Kernel, Unsloth, and the Triton tutorials run through the same referee under two conditions: *fixed-schedule* (raw extracted `@triton.jit` kernels under our wrappers; schedule pinned) and *library-API* (Liger called via its own public `Function` interface, exactly as shipped). Ours is faster on `swiglu/rmsnorm/relu2/geglu` in both conditions; softmax and layernorm are ties-within-noise against the best fixed-schedule expert. Two honesty notes. First, **we retract our own earlier claim** that 5 of 11 expert kernels were non-robust to odd shapes: under the library-API condition all 5 Liger recipes pass our adversarial sweep — the failures were artifacts of *our* extraction wrappers. Second, the forward-only cross-entropy comparison is *unfair to Liger* (their CE computes gradients in the forward by design) and is excluded from any headline.

5.3 End-to-end composition

A Qwen-style MLP block (`add_rmsnorm` $\rightarrow$ gate/up GEMMs $\rightarrow$ `swiglu` $\rightarrow$ down GEMM) runs three ways on identical weights; output verified against eager (max err $2.9\text{e-}03$) before timing. With trained kernels (NVIDIA H200): $1.143\times$ vs eager and $0.993\times$ vs whole-block compile-MA — the compile-MA comparison is *a tie, reported as a tie*: GEMMs are 81% of the block and identical across paths, bounding the achievable win (Amdahl), though the non-GEMM sub-path itself is $10.88\times$ faster. (On an RTX 4090 with gold seeds the same block runs $1.085\times$ / $1.302\times$, where the compiler is weaker.) The single-op wins survive composition; they cannot and do not claim block-level miracles.

5.4 Teaching 37 unseen operators in one run

Resuming the RL loop on 37 operators absent from all prior training (8 new epilogues $\times$ norm/residual variants, plus softcap-softmax, Gemma-RMSNorm, GLU, interleaved RoPE, fused cross-entropy), with the templated-mutation arm disabled (the model wrote every candidate): **37/37 validated correct and beat `torch.compile` max-autotune on fresh exit measurement** ($1.16\times$ – $2.09\times$), 35/37 winning kernels model-authored, 617/888 sampled kernels verified (0.695), 101 archive lead-takes, explore-arm contribution **0**. The model beat its own strongest gold seed on fused cross-entropy ($2.085\times$ vs `torch.compile` max-autotune, model-authored). Learning is visible *inside* the run: the `softplus/mish` family — which requires an overflow-guard idiom (`softplus(x) = x` for $x > 20$, else $\log(1+e^x)$) — verified 3/1/1/1 per round on first encounter and 8/8/8/8 on the second pass, after self-distillation on the first pass’s verified winners (Fig. 3).

arm	removes	valid-rate	beat MA	geomean vs MA
control	—	1.000	8/8	1.338
no-feedback	referee errors in prompt	0.995	8/8	1.309
distill-only	policy-gradient term	0.948	8/8	1.361
no-learn	all weight updates	0.969	8/8	1.302

Table 2: Ablations on familiar operators (single seed; the ordering among 1.30–1.36 is suggestive, not established). The no-learn row’s headline was recovered from its completed report before a concurrent-write race overwrote the stored copy; the race is documented and fixed in the artifact, and the arm’s kernels remain archived for re-validation.

5.5 The invention probe, and the repair of the loss regime

We then aimed the trained model at seven problems chosen to be *foreign*: `cumsum` (a prefix *scan* — carry propagation, a different parallel algorithm class from every reduction in the suite), `entropy` and `kl.div` (coupled double reductions over logits), and four operators pinned to the 16384×2048 loss regime identified by the shape grid. Pre-registered prediction (ours, in writing): the regime requires split-row/multi-row schedules.

Outcome: 7/7 validated and beat `torch.compile` max-autotune; 5/7 model-authored; validity on foreign problems 0.497 (vs ~ 0.69 on familiar families — an honest difficulty gradient). Reading the kernels, not just the scores (Fig. 4):

- **The human diagnosis was wrong.** The winning kernels are plain *whole-row single-block* schedules; the losses were caused by looped templates tuned at $N=4096$, not by row-per-program parallelism. The worst published loss cell ($0.69\times$, `add_layernorm_sigmoid`) became a $1.222\times$ win; `layernorm_gelu` went $0.82\times \rightarrow 1.450\times$; softmax reached $1.803\times$ in-regime.
- **`cumsum` is invention-lite, labeled precisely.** The model did not re-derive carry propagation; it recognized that a whole row fits one block and emitted a loop-free `tl.cumsum` kernel — beating our hand-written carry-loop gold by $\sim 47\%$ ($1.292\times$ vs `torch.compile` max-autotune, $1.896\times$ vs `compile`), correct across the full sweep. Primitive selection and schedule simplification, verified — not a new parallel algorithm.
- **The capability boundary, stated:** on `entropy` and `kl.div` the model produced many verified kernels but never beat the hand-written two/three-pass golds (which themselves stand at $1.477\times$ and $1.225\times$ vs `torch.compile` max-autotune). Authorship there remains human.

6 Ablations: where does the value come from?

Four arms, identical except one ingredient (8 familiar ops, same SFT starting adapter, 24 rounds, group 8, single seed — labeled): Table 2, Fig. 5.

Three observations, each contrary to at least one of our own pre-registered predictions: **(1)** On familiar operators, the referee plus best-of- N sampling from a frozen competent model captures 97% of the full loop’s geomean — *search, not learning, is the dominant force where competence exists*. **(2)** The distill-only arm matched-or-beat control; the group-relative policy-gradient term earned nothing here. Self-distillation on verified winners appears to be the load-bearing learning ingredient. **(3)** Feedback’s effect is unmeasurable at the familiar-op validity ceiling — its value, like learning’s, concentrates on foreign problems (the 0.497 vs 0.69 validity gap and Fig. 3 are the controlled contrast). A historical datum completes the picture: a 4.5-hour *continue-SFT* on new operators stalled and degraded proven operators and was stopped, while RL-from-verifier taught the same operators cleanly — corpus imitation is not how the new capabilities arrived.

7 The falsification ledger

Because the referee is cheaper to consult than to argue with, we adopted the practice of writing predictions down *before* runs. Across the campaign the referee falsified 9 confident human beliefs, including: the loss-regime diagnosis (§5.5); the authors’ hand-written scan kernel (−47%); the published expert-robustness claim (§5.2, retracted); the superiority of continue-SFT for new operators; the ablation ranking (§6); the adequacy of a subtle anti-memoization perturbation (§3.3 — caught by the system’s own negative control); the verifiability of an innocuous-looking operator (`l1norm`); an early benchmark that manufactured compiler ties (§3.2d); and the evidentiary value of all-green logs without durable artifacts (which forced a re-run of our own stability gate under a written provenance note). We highlight this not as housekeeping but as the paper’s second thesis: **a system with an honest referee does not require its builders to be right — and will routinely demonstrate that they are not.** The ledger, with open pre-registered predictions, ships in the artifact.

8 Honest bounds

(1) These are scheduling wins on bandwidth-bound fusion operators — no claims against cuBLAS/FlashAttention-class kernels, and no novel algorithm classes (the scan result is primitive selection). (2) All speedups are vs `torch.compile` max-autotune on specific hardware (H200; spot-confirmed on RTX 4090); ratios are hardware-dependent even when reproducible. (3) Ablations are single-seed and labeled. (4) KernelBench cross-evaluation is out-of-scope-by-construction for our specialist (foreign format, generalist operators); the recorded honest probe scored 0/1 and we declined prompt-bridging shortcuts that would manufacture a non-comparable number. (5) The loop requires a domain with a cheap, un-gameable oracle and a measured quality signal; domains with judged rewards are explicitly outside the recipe, and the framework’s failure modes (expensive oracles, adversarial environments, Goodhart on the metric) are discussed in the artifact.

9 Reproducibility

Everything regenerates from commands: the harness self-test (`harness.py`; 36 cases), the gates (`rebench_stability.py`, `rebench_shapes.py`), training (`modal_app.py` entrypoints), and this paper’s every number (`paper/make_numbers.py`, which fails the build on cross-report inconsistency) and figure (`paper/make_figures.py`). The artifact includes the 76 kernels, adapters (2B and 27B), the verified corpus, raw five-sample stability data, per-cell grid data, run transcripts where retrievable (with an honest note on which are windows), and the falsification ledger.

10 Conclusion

OUROBOROS set out to show that the reward channel of an RL-for-code system can be hardened the way security-critical code is hardened — with its exploits implemented, committed, and continuously rejected — and that a commodity open model trained against such a referee does real engineering: 76 reproducible compiler-beating kernels, its own weak regime repaired, its builders corrected nine times. The model was swapped once (2B→27B) and could be swapped again; nothing in the result depends on its identity. What cannot be swapped out is the referee. In domains where verification is cheap and honest, that inversion — *the verifier is the asset; the model is a commodity* — is, we believe, the durable lesson, and the one most worth carrying to the next domain.

Acknowledgments

Engineering, experiments, and drafting were carried out with Claude (Anthropic) operating the OUROBOROS pipeline end-to-end; all claims trace to harness-emitted artifacts rather than to either human or model assertion. Compute: Modal (H200/L4) and a local RTX 4090.

References

- [1] Sakana AI. *The AI CUDA Engineer*. Technical report and subsequent public correction, February 2025.
- [2] A. Ouyang, S. Guo, et al. *KernelBench: Can LLMs Write Efficient GPU Kernels?* arXiv preprint, 2025.
- [3] C. Baronio, P. Marsella, et al. *Kevin: Multi-Turn RL for Generating CUDA Kernels*. arXiv preprint, 2025.
- [4] S. Li et al. *AutoTriton: Automatic Triton Programming with Reinforcement Learning in LLMs*. arXiv preprint, 2025.
- [5] X. Li et al. *CUDA-L1: Improving CUDA Optimization via Contrastive Reinforcement Learning*. arXiv preprint, 2025.
- [6] A. Fawzi et al. *Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning*. Nature 610, 2022.
- [7] D. Mankowitz et al. *Faster Sorting Algorithms Discovered Using Deep Reinforcement Learning*. Nature 618, 2023.
- [8] E. Schkufza, R. Sharma, A. Aiken. *Stochastic Superoptimization*. ASPLOS, 2013.
- [9] Z. Shao et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models*. arXiv:2402.03300, 2024.
- [10] N. Lambert et al. *Tulu 3: Pushing Frontiers in Open Language Model Post-Training*. arXiv preprint, 2024.
- [11] J. Ansel et al. *PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation*. ASPLOS, 2024.
- [12] P. Tillet, H. T. Kung, D. Cox. *Triton: An Intermediate Language and Compiler for Tiled Neural Network Computations*. MAPL, 2019.
- [13] P.-L. Hsu et al. *Liger Kernel: Efficient Triton Kernels for LLM Training*. arXiv preprint, 2024.

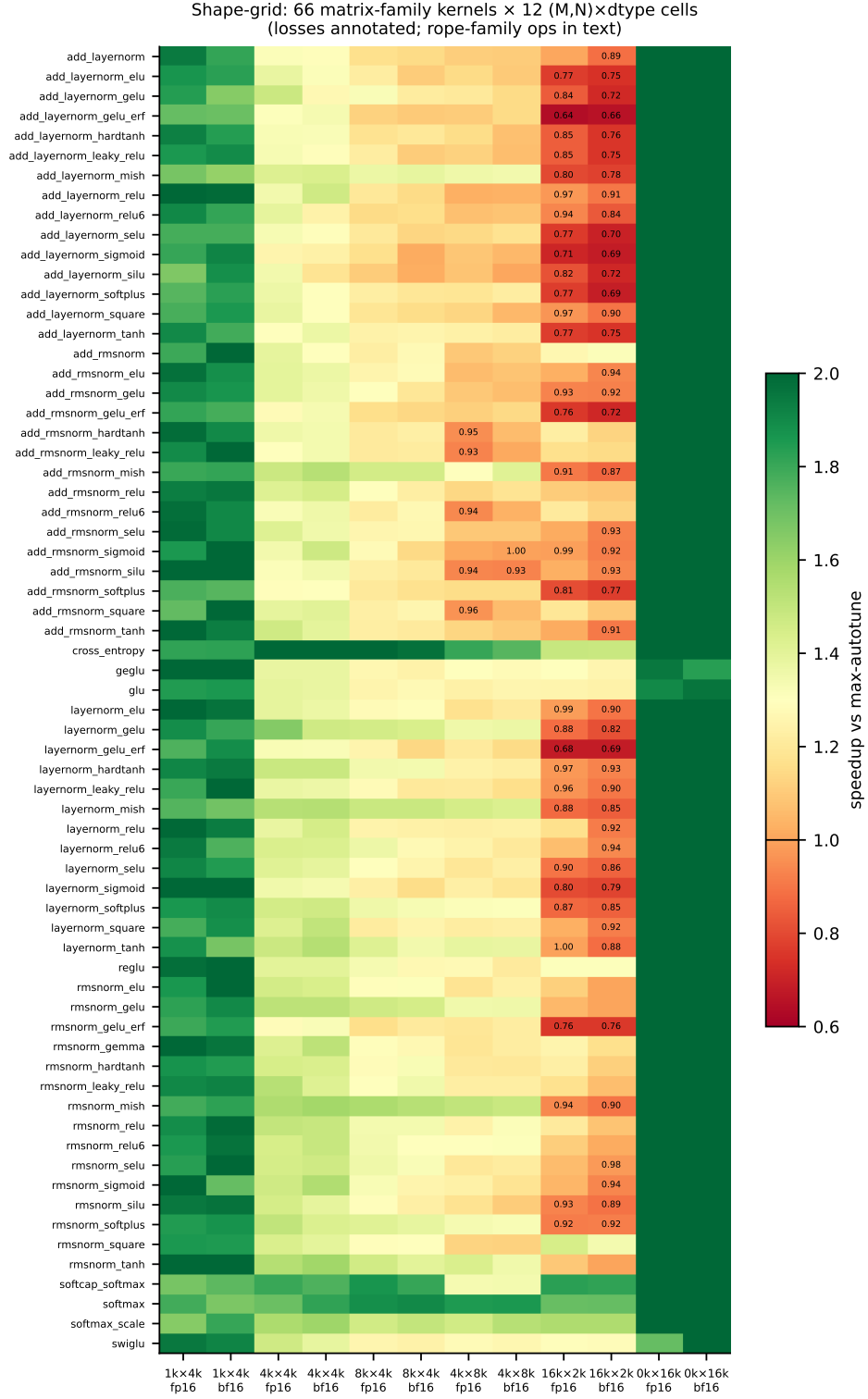


Figure 1: The shape grid. Green: beats `torch.compile` max-autotune (recompiled per cell); annotated red cells: losses, printed rather than hidden. The loss region is one regime, not noise.

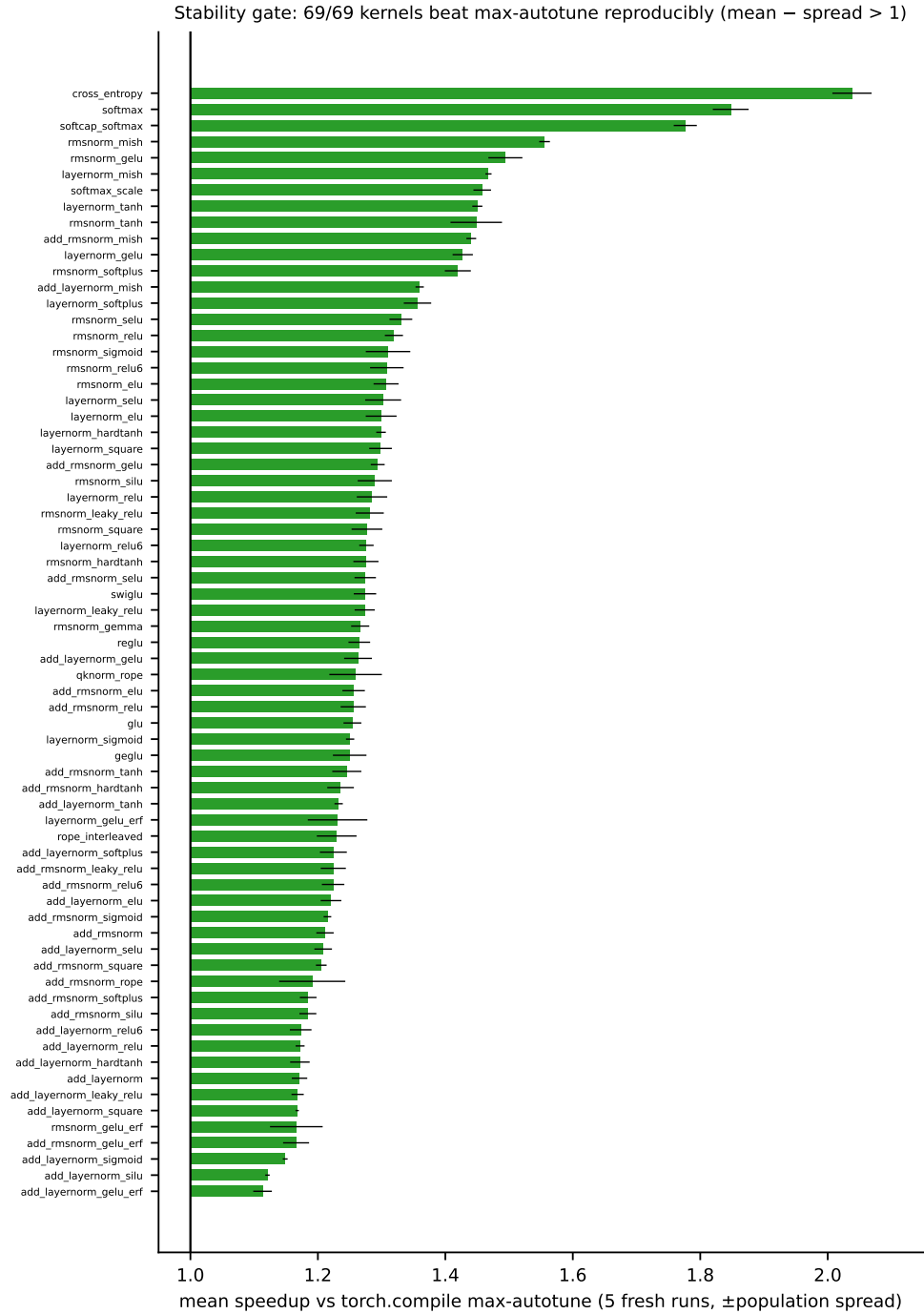


Figure 2: The 5 $\times$ stability gate over all 69 kernels: mean $\pm$ spread vs `torch.compile` max-autotune. Raw five-sample data for every bar ships in the repository.

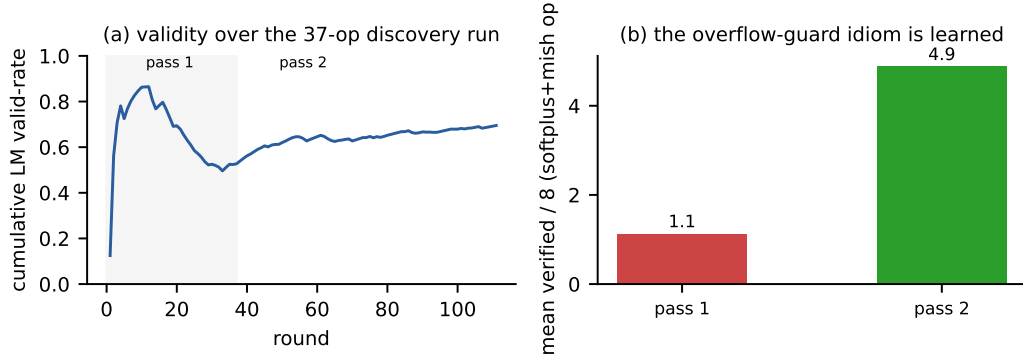


Figure 3: Learning where it matters: (a) cumulative validity over the 37-unseen-op run dips as harder epilogue families arrive, then recovers as distilled idioms generalize; (b) the overflow-guard idiom, quantified.

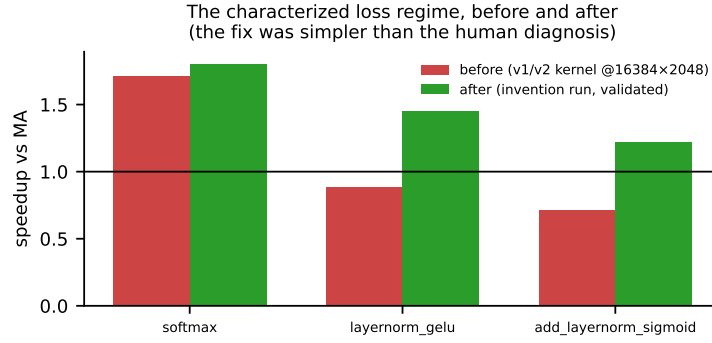


Figure 4: The characterized loss regime (16384×2048 , fp16), before and after the invention run. The repair used a *simpler* schedule family than the authors predicted in writing.

Ablations: 8 familiar ops, 24 rounds, single seed — the GRPO term earns nothing on familiar ops

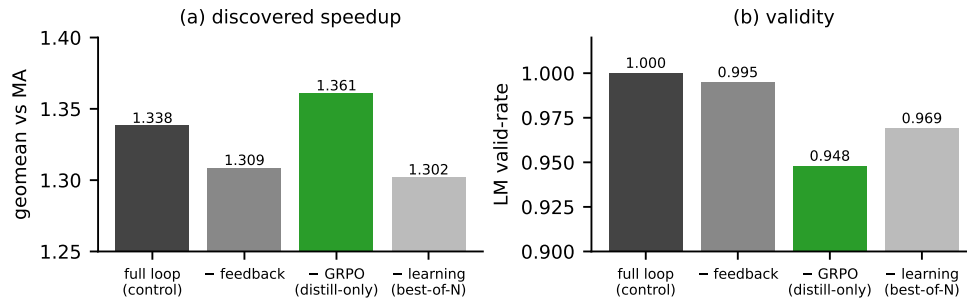


Figure 5: Ablation arms. Removing the policy-gradient term does not hurt discovered speedups on familiar ops; removing *all* learning costs only a few percent there.